

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 1(a)(i)   | <p>measure, length / <b>A</b> / AW, (on image) <b>and</b> divide by, magnification / 5000 ;<br/> <b>A</b> size for length to convert from mm to <math>\mu\text{m}</math> multiply by 1000 ;<br/> <b>A</b> cm to <math>\mu\text{m}</math>, multiply by 10 000<br/> <b>A</b> if shown as part of the actual calculation</p> <p><i>if no marks gained allow one mark for correct formula</i></p> $\text{actual length} = \frac{\text{length of, } \mathbf{A} / \text{organelle / mitochondrion / image}}{\text{magnification / 5000}}$ <p><b>bod</b> <math>A = I / M</math> <b>or</b> <math>I = A \times M</math> <b>or</b> <math>M = I / A</math> <b>or</b> magnification triangle</p>                                                                                          | 2 |
| 1(a)(ii)  | <p>nucleus / nuclear envelope<br/> nucleolus<br/> cell surface membrane<br/> Golgi body<br/> <b>A</b> Golgi, apparatus / complex<br/> <b>bod</b> lysosome(s)<br/> <b>I</b> cytoplasm / mitochondria / vesicles</p> <p>} any <b>two</b> for one mark ;</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1 |
| 1(a)(iii) | <p>any <b>two</b> from:<br/> <b>A</b> ora for TEM</p> <p>1 light microscope has, low(er) / poor(er), resolution (than EM) ;<br/> 2 <i>idea of</i> light microscope has a lower ability to distinguish between points ;<br/> <b>A</b> definition of resolution<br/> 3 structures closer together than, <math>0.2 \mu\text{m}</math> / 200 nm, are not visible as separate<br/> <b>or</b><br/> a structure smaller than, <math>0.2 \mu\text{m}</math> / 200 nm, is not visible<br/> <b>A</b> range / within range, of, <math>0.1 \mu\text{m}</math> – <math>0.3 \mu\text{m}</math> / 100 – 300 nm<br/> <b>or</b><br/> some structures are smaller than the resolution of the light microscope ;<br/> <b>A</b> some structures are smaller than half the wavelength of light</p> | 2 |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1(b)     | <p>any <b>three</b> from:</p> <ol style="list-style-type: none"> <li>1 nucleus is site of, genes coding for enzymes / transcription / production of messenger RNA ;<br/><b>A</b> mRNA</li> <li>2 (80S) ribosomes / <u>rough endoplasmic reticulum</u>, for, protein / enzyme, synthesis / production ;<br/><b>I</b> RER<br/><b>A</b> for, translation / assembling of amino acids / polypeptide formation<br/><b>R</b> if 70S ribosomes</li> <li>3 rough endoplasmic reticulum detail ; <b>ecf</b> on abbreviation (RER) in mp2<br/>e.g.<br/>for post-translational modification / modification of protein(s) / protein folding / AW<br/>for production of transport vesicles <i>context is to Golgi body</i></li> <li>4 Golgi body, for modification of protein / packaging protein into (Golgi / secretory) vesicles ;<br/><b>A</b> vacuoles <i>for vesicles</i><br/><b>A</b> glycosylation / processing / described, for modification<br/><b>I</b> any qualification of protein, e.g. defective</li> <li>5 Golgi / secretory, vesicles for, transport to cell (surface) membrane / exocytosis / fusion with cell (surface) membrane ;</li> <li>6 mitochondria provide, energy / ATP (for protein synthesis / movement of vesicles / exocytosis / amino acid activation) ;</li> <li>7 AVP ; e.g. nucleolus for production of, rRNA / ribosomal subunits / ribosomes</li> </ol> | 3     |